

Week 1: Jumping pegs

Facilitator guide for the three adults (K–1, grades 2–3, grades 4–5)

This guide is one document for the three adults. Each table's section is self-contained: order and timing, complete answers and arguments for every problem, hints in the order to give them, what to do when the table stalls or finishes early. The student pages carry only the problems, the boards and the recording space; every explanation, script and teaching note is here. On the student pages a grey circle means “a counter starts here”, a thick line is the line the army must cross, and a star is the target square. Strip positions in this guide are written left to right with X for a counter and . for an empty square, so XXX. is a strip of four with the last square empty.

1. The session at a glance

- 0:00–0:04 Arrival and free handling. Each table has its first page and its cup of counters. Children may put counters anywhere and jump; correct only diagonal jumps, slides, and jumps over an empty square.
- 0:04–0:07 Whole-group launch at the big board (script in Section 3), then children go to their tables.
- 0:07–0:50 Table work (43 minutes). Orders and timings in Sections 4–6.
- 0:50–0:58 Share (Section 7).
- 0:58–1:00 Counters back in the cups, pages home with the children.

Adults: organizer (math PhD) at grades 4–5; second mathematician at grades 2–3; parent at K–1. The K–1 table needs reading aloud and scribing above all, and every argument the table needs is written out in Section 4 so that nothing has to be worked out during the hour. Any other assignment works; each table's section is self-contained. The two mathematicians should each drop by the K–1 table for a minute during Problems 9 and 12 (the two arguments) if their own table is in the middle of a long problem.

The one rule for the whole hour: a counter may jump over the counter right next to it (left, right, up or down, never diagonally) into the empty square immediately beyond; the counter jumped over is removed. No sliding, no jumping over an empty square, no jumping over two counters.

2. Materials and preparation

Materials (counts):

- Pennies: 100. Cup of 40 for grades 2–3 (four children; the row-3 army needs 8 each), cup of 60 for grades 4–5 (three children; the row-4 army needs 20 each).
- Checkers: 24, one full set, for K–1 (8 per child, four of each colour, so that a checker of the other colour is at hand for the reserve “which checker is the last one”); a second set stays in the bag as spares.
- Student pages: K–1 packet $\times$ 3, grades 2–3 packet $\times$ 4, grades 4–5 packet $\times$ 3 (one per child; the strips and boards are played on directly). Facilitator guide $\times$ 3.
- Pencils: 12. Crayons: 3 (K–1 Problem 11 colours a square). Two erasers.

- Three strips of eight 1.25-in squares cut from a printed 1.25-in grid, for K–1 Problem 6 (one per child).
- Launch and share board: one printed 1-in grid at least 9 squares wide and 8 squares tall (portrait Letter works). Draw a thick line with a marker between the 4th and 5th rows from the bottom (four rows below the line, four above). Use a small sticky note as the star; it is moved to row 1 for the launch and to rows 2, 3, 4 during the share. Two spare pennies for the launch.
- Four spare printed 1-in grids and a marker, for reserve boards (outline a 3-by-4 or 3-by-3 board when needed). Ten sheets of scrap paper.

Preparation (15 minutes the day before): print and staple the packets; cut the three strips of eight; prepare the launch board; sort the cups; each adult reads their table's section (4, 5 or 6) once through and plays the first three problems with real counters.

3. Whole-group launch (2 minutes)

Everyone stands around the launch board (star on row 1, two pennies in the star's column in rows 0 and -1 , nothing else).

Script. "This is the line. These two counters start below it. Here is the only rule we use all hour: a counter may jump over the counter right next to it, left, right, up or down, never diagonally, into the empty square right behind it, and the counter it jumped over comes off. [Invite a K–1 child to make the jump: the lower penny over the upper one onto the star.] One counter is over the line. [Remove the jumped counter, leave the one on the star, then put a single penny back below the line alone.] Can one counter alone ever go anywhere? [Let them answer: no, nothing to jump over.] So it took two counters to get one over the line. At your tables you will work out how far an army can get, how few counters you can leave on a strip, and the question that matters most: how you can be sure when something cannot be done. Go."

Do not state any answers beyond "two counters get one row over". Do not mention the golden ratio; it is for the share and the mathematicians.

4. K–1 table (parent; three children)

Order and timing (table work runs from 0:07 to 0:50, 43 minutes; the pages hold 67 minutes): Problems 1–4 by 0:24, Problems 5–6 by 0:33, Problems 7–10 by 0:50 (the army; these are what the table shows at the share). Problems 11–15 are for a fast table or a child who races ahead, and the reserves in Section 4.4 come after those. Whatever happens, start Problem 7 by 0:33: if the table is still on Problem 4 at 0:30, finish recording it and skip Problems 5–6 (they can be done at the end). The child places the checkers and jumps; the adult reads each problem aloud, scribes the dots, tallies and numbers when asked, and asks the "how do you know" questions listed below only after the recording is done.

Rows and columns of the army boards are named here from the thick line: the row just above the line is row 1, the next row 2; the row just below the line is row 0, then -1 , -2 , -3 . Columns are counted from the star's column (0), positive to the right.

4.1 Answers and arguments

Problem 1. Strip A (checkers on 1 and 2): one jump, 1 over 2 into 3, landing on square 3; one checker left. Strip B (checkers on 1 and 3): no jump. The middle square is empty, a checker may only jump over a checker, and neither checker has a neighbour. Strip C: one jump, 3 over 2 into 1, landing on square 1. Marks beside A and C.

Problem 2. Strip D (XXX.): the only possible jump is 2 over 3 into 4. Then X..X, no jump left. One jump, two checkers left (squares 1 and 4). There is never a choice.

Problem 3. Strip E (X.XX): the only first jump is 4 over 3 into 2 (XX..), then the only jump is 1 over 2 into 3 (..X.). Two jumps, one checker left, on square 3. Strip E allowed more jumps. Same three checkers, same strip, the empty square moved by one, and the ending changed from two left to one left.

Problem 4. Every play on a strip of five leaves two checkers. Strip F (XXXX.): forced, 3 over 4 into 5, then 1 over 2 into 3, stuck at ..X.X. Strip G (XXX.X): forced, 2 over 3 into 4, then 5 over 4 into 3, stuck at X.X... Strip H (XX.XX): two possible first jumps. After 1 over 2 into 3 (..XXX) the only jump is 4 over 3 into 2, stuck at .X..X. After 5 over 4 into 3 (XXX..) the only jump is 2 over 3 into 4, stuck at X..X.. The mark goes beside H. Why no strip of five can get down to one: on strip H there are only these five positions in the whole game (the start, two positions after the first jump, two stuck endings) and both endings have two checkers. A child can show this by playing H both ways and pointing out that F and G never offered a choice.

Problem 5. Strip J (XXXXX.): forced, 4 over 5 into 6, then 2 over 3 into 4, stuck at X..X.X, three left. Strip K (XXXXXX.): forced, 5 over 6 into 7, 3 over 4 into 5, 1 over 2 into 3, stuck at ..X.X.X, three left.

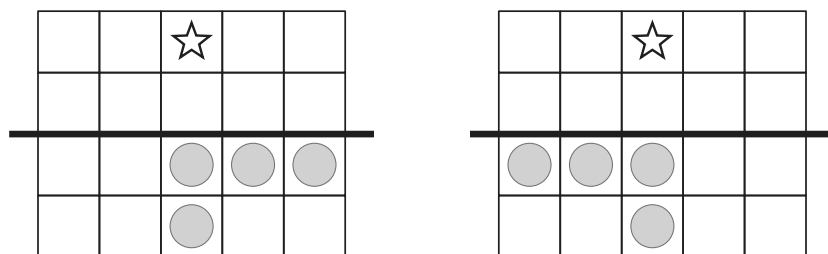
Problem 6. Endings with the empty square at the end: 4 squares X..X (2 left), 5 squares ..X.X (2), 6 squares X..X.X (3), 7 squares .X.X.X (3), 8 squares X..X.X.X (4). What is the same: the checkers left sit on every second square counted from the right end, and there is never a choice. The strip of eight: 6 over 7 into 8, 4 over 5 into 6, 2 over 3 into 4, stuck at X..X.X.X, four left. Reason a child can give: the only checker that can move is the one two squares from the empty square; it jumps into the hole and leaves a new hole two squares further left; then the same thing happens again, until the hole reaches the left end. Ask: "Where would they be on a strip of nine?" (..X.X.X.X, four left.)

Problem 7. Two checkers: the lower one jumps over the upper one onto the star. One checker alone never jumps: there is nothing to jump over.

Problem 8. Four checkers: a pair stacked under each star (rows 0 and -1 in each star's column); each pair does the jump of Problem 7. Fewer than four is impossible (see Problem 9).

Problem 9. Six checkers, a pair under each star. Four stars would need eight. The argument, in the children's terms: every checker that ends on a star got there by jumping, because checkers only move by jumping; every jump throws one checker off; so for every star there is one checker on it and at least one checker thrown away for it, at least two per star. Have a child show it with checkers: put a pair under one star, jump, and point to the thrown-away checker; "this one was used up for that star". Sequence for the table: 1 star 2, 2 stars 4, 3 stars 6, 4 stars 8.

Problem 10. The L reaches the star in three jumps: (1) the lowest checker jumps up over the one above it, landing just above the line in the star's column; (2) the rightmost checker in the row below the line jumps left over its neighbour into the star's column; (3) that checker jumps up over the one above the line onto the star. On the small boards: arrow up, arrow left, arrow up.



The L of Problem 10 (board D) and its mirror image: the only two four-checker armies that reach a star two rows up (Problem 13).

Problem 11. The column of four reaches only the row just above the line, and on board E the play is forced: the second checker from the top jumps up over the top one to row 1 (the only jump available); then the bottom checker jumps up over the one above it into row -1 (again the only jump); then nothing can jump (the two remaining checkers are two squares apart with an empty square between them). Colour the square just above the line. Board D reached the star, board E did not: the same four checkers, a different shape.

Problem 12. Three checkers never reach a star two rows up (checked by complete search over every placement). Argument for the adult, to be given in pieces: with three checkers you can make at most two jumps, because each jump throws one checker off and the checker on the star must remain. To land on the star, the last jump comes from the square just below the line over a checker in the row just above the line (a sideways jump inside row 2 would need two checkers already up in row 2, which is even more work). So just before the last jump we need one checker in row 1 and one checker in row 0 under it. The checker in row 1 got there by jumping up over the checker that was in row 0, and that jump threw that row-0 checker away. So someone else must refill row 0 before the last jump: that is a third jump, and three jumps need four checkers. The L does exactly this: up, then refill from the side, then up.

Problem 13. The two ways: the L of Problem 10 (three checkers in row 0 starting at the star's column and going right, plus one checker under the first) and its mirror image (going left), drawn above. Complete search shows these are the only two placements of four checkers that work. If a child offers a shape with the checkers elsewhere, let them test it; it will fail.

Problem 14. Middle empty: no jump at all. A jump needs an empty square two steps away in a straight line with a checker in between; the only empty square is the middle, and every checker is either next to it (the side squares) or diagonal from it (the corners), never two steps away in a straight line. Corner empty (bottom-left): exactly two first jumps, the top-left checker down over the left-middle one, or the bottom-right checker left over the bottom-middle one.

Problem 15. The fewest is four; every play ends with four or five checkers (16 positions in the whole game). A play that leaves four: top-left down over left-middle into the corner; right-middle left over the centre into left-middle; corner up over left-middle into top-left; bottom-right left over bottom-middle into the corner; stuck with the top row full and one checker in the bottom-left corner. The ending with five is an L (the whole left column and the whole bottom row), reached for instance by top-left down into the corner, top-right left over top-middle into top-left, right-middle left over the centre into left-middle. The three endings with four are: top row plus the bottom-left corner; right column plus the bottom-left corner; and the four checkers top-left, top-right, centre, bottom-right. One checker is impossible here; a child can say "we tried every first jump and every way after it" (there are only 16 positions).

4.2 Hints, in order

- Problem 1: “Which checker has a checker right next to it? Where is the empty square behind that one?” Then demonstrate one jump on strip A and let the child do B and C.
- Problems 2–4: “Find every checker that could jump right now. How many are there?” (On D, E, F, G the answer is always one; on H it is two.) If a child slides or jumps over a gap, reset and restate the rule.
- Problem 6: “Look at the right-hand end of each small strip. Where is the first checker? The next one?” For the prediction: “Do it in your head: which checker jumps first on the strip of eight?”
- Problem 8: if a child puts three or four checkers under one star: “That star is done. What did the other star need?” If they try a row of checkers along the line: “Which way must the checker jump to cross the line? What must be under it?”
- Problem 9: “Show me the checker that got thrown away for this star. And for that one.”
- Problem 10: “Which jump crosses the line first? Now which checker can jump into the middle column?”
- Problem 12: after two or three failed armies, “How many jumps can three checkers ever make?” then “What has to be under the star just before the last jump?”
- Problem 14: “Point to the empty square. Is any checker two steps away from it in a straight line?”
- Problem 15: “Try starting with the other first jump. Which checker could you save at the end?”

4.3 If the table stalls

Reading and placing take most of the time at this table. Keep the rhythm: the child moves, the adult scribes. If a child cannot find any jump on a strip, do the first jump with the child's hand and let them find the next. Never leave a problem unrecorded; the dots on the small strips are what the share uses. If Problems 12–13 stall, stop after five minutes, show the L and its mirror, and go on to Problem 14, which is a fresh start.

4.4 If the table finishes early (reserves, no printing needed)

- R1. The count that never changes. On strip J put checkers on squares 1, 2, 4, 5, 6 (XX.XXX). Play twice with different first jumps. The two endings are X.X.X (2 jumps, 3 left) and .X.X. (3 jumps, 2 left); jumps plus checkers left is 5 both times, because each jump throws off exactly one checker.
- R2. Which checker is the last one. On strip J set X.XXXX (checkers on 1, 3, 4, 5, 6) with a checker of the other colour on square 1. Every play ends with one checker (7 positions in the game). The checker from square 1 can be the survivor (it ends on square 5: 4 over 3 into 2, 1 over 2 into 3, 6 over 5 into 4, 3 over 4 into 5) and the checker from square 6 can be the survivor (it ends on square 2: same start, then 4 over 3 into 2 as the last jump). A checker that starts on 3, 4 or 5 never survives.
- R3. Count the first jumps. On strip H (XX.XX): 2 first jumps. On X.XXXX: 1. On board J: 2. Ask the child to point to each.
- R4. Last jump wins, on board J (corner empty). Two players take turns making one jump; whoever makes the last possible jump wins. The starter wins by jumping the top-left checker down into the

corner; whichever of the two available replies the opponent makes (right-middle into left-middle, or top-right into top-left), the starter makes the other one and no jump is left.

5. Grades 2–3 table (second mathematician; four children)

Order and timing (table work runs from 0:07 to 0:50, 43 minutes; the pages hold 63 minutes): Problems 1–2 by 0:16, Problem 3 by 0:24, Problem 5 by 0:32, Problems 6–7 by 0:42, Problem 8 by 0:50; Problem 9 is for the pair that gets there and for the share (the adult places the army at the share in any case). Problem 4 is done only if the table is ahead of these times; otherwise it is the first thing to do after Problem 8 or at the end. Whatever happens, start Problem 6 by 0:36 so that the army problems are reached. Let the children work in two pairs on Problems 3, 7, 8 and 9; each child still records on their own page.

Rows of the army boards are named from the thick line (row 1 just above it, row 0 just below it, then -1 , -2 , ...) and columns from the star's column (0), positive to the right.

5.1 Answers and arguments

Problem 1. Strip A (XXX.): the only jump is 2 over 3 into 4; stuck at X. .X; 1 jump, 2 left. Strip B (X.XX): 4 over 3 into 2, then 1 over 2 into 3; stuck at . .X.; 2 jumps, 1 left. Same three pennies; moving the hole by one square changes the ending.

Problem 2. Strip C (XXXX.): forced, 3 over 4 into 5, 1 over 2 into 3, stuck . .X.X, 2 left. Strip D (XXX.X): forced, 2 over 3 into 4, 5 over 4 into 3, stuck X.X. ., 2 left. Strip E (XX.XX): first jump 1 over 2 into 3, then 4 over 3 into 2, stuck .X. .X; or first jump 5 over 4 into 3, then 2 over 3 into 4, stuck X. .X.; 2 left either way. Only strip B is circled. On a strip of five no hole position allows one penny (the other hole positions are mirror images of C and D).

Problem 3. Strip F (X.XXXX, hole at 2): the first jump is forced (4 over 3 into 2, giving XX. .XX), then there are choices, but every play ends with exactly one penny (the game has 7 positions; the two endings are .X. . . . andX.). The last penny sits on square 2 or square 5, never anywhere else. Example: 4 over 3 into 2, 1 over 2 into 3, 6 over 5 into 4, then 3 over 4 into 5 (ends on 5) or 4 over 3 into 2 (ends on 2). Strip G (XX.XXX, hole at 3): two first jumps. Starting 1 over 2 into 3, the rest is forced: 4 over 3 into 2, 6 over 5 into 4, stuck .X.X. ., 2 left. Starting 5 over 4 into 3, the rest is forced: 2 over 3 into 4, stuck X. .X.X, 3 left. So the fewest is 2, and the first jump decides it. Strip H (XXXXX., hole at 6): forced, 4 over 5 into 6, 2 over 3 into 4, stuck X. .X.X, 3 left. Compare: hole at 2 always 1; hole at 3, 2 or 3; hole at 6 always 3.

Problem 4. Strip J (XXX.XXXX): the endings are .X. andX. . . (1 penny, 6 jumps), . .X. . .X. (2 pennies, 5 jumps) and X.X.X. .X (4 pennies, 3 jumps). The last column is 7 every time. Reason: every jump throws off exactly one penny, so the number of jumps equals the number of pennies removed, and removed plus left is the starting seven. Sample plays: to 1 penny: 6 over 5 into 4, 3 over 4 into 5, 1 over 2 into 3, 8 over 7 into 6, 6 over 5 into 4, then 3 over 4 into 5 (ends on 5) or 4 over 3 into 2 (ends on 2). To 2 pennies: 6 over 5 into 4, 8 over 7 into 6, 3 over 4 into 5, 1 over 2 into 3, 5 over 6 into 7. To 4 pennies: 2 over 3 into 4, then the forced 5 over 4 into 3 and 7 over 6 into 5, and no jump is left. Ending with exactly 3 is impossible (there are 16 positions in the game and no stuck position has 3).

Problem 5. The map of XX.XX has 5 positions and 4 arrows: XX.XX $\rightarrow$. .XXX and XXX. .; . .XXX $\rightarrow$.X. .X (stuck); XXX. . $\rightarrow$ X. .X. (stuck). Both stuck positions have 2 pennies. Because the map contains every position that can ever be reached, it is a proof that one penny is impossible on this strip: there is no way

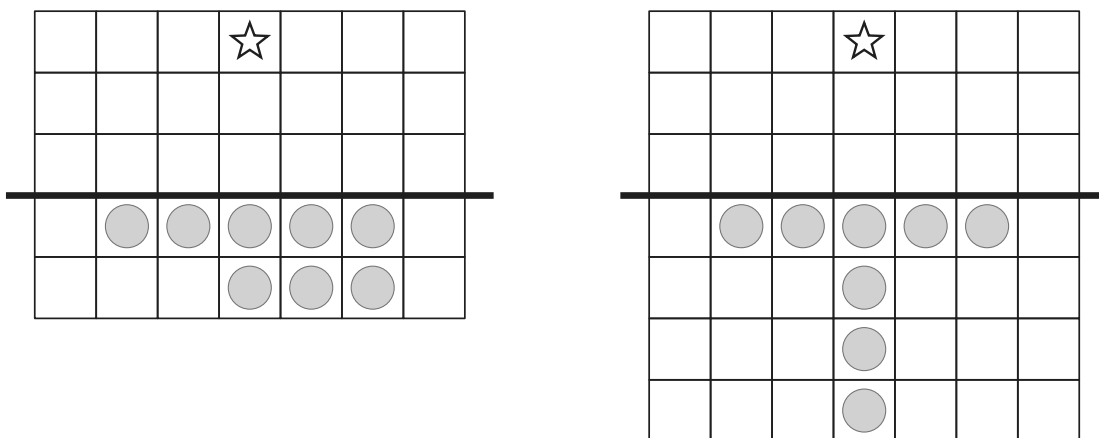
to play that is not on the map. Ask: “Could there be a play we missed?” The answer is no, because at every position we drew every possible jump.

Problem 6. Always exactly one square past the line, and the play is forced. K (block of 3): 2 over 3 into 4, then nothing; farthest square 4. L (block of 4): 3 over 4 into 5, 1 over 2 into 3, then nothing; farthest square 5. M (block of 5): 4 over 5 into 6, 2 over 3 into 4; farthest square 6. Number boxes: 1, 1, 1.

Problem 7. No army on squares 1 to 6 ever gets a penny onto square 8; the farthest any penny ever reaches is square 7 (all 63 armies checked by computer). Gaps do not help. The argument for the table, with prices (this is the grades 4–5 Problem 4; the adult leads it here with pencil and paper): write 1, 1, 2, 3, 5, 8 under squares 1 to 6 and 13, 21 under squares 7 and 8. From square 3 on, every price is the sum of the two prices before it. A jump to the right removes pennies from two neighbouring squares and puts one on the next square, whose price is exactly the sum of the two: the total price of the pennies on the strip does not change. A jump to the left lands on a square that costs less than the two it used together: the total goes down. So the total never goes up. The whole army on squares 1 to 6 is worth $1 + 1 + 2 + 3 + 5 + 8 = 20$, and any smaller army is worth less. A penny sitting on square 8 is worth 21 by itself. So no penny ever gets there. Third graders can follow this if the adult does the adding with them on one of their small strips. A shorter picture, less rigorous but good for the table: to land on 8 you need pennies on 6 and 7; the penny on 7 came from 5 over 6, which emptied 6; to refill 6 you need 4 and 5 full again; refilling 5 needs 3 and 4; and so on, each refill needing a refill two squares further left, until the chain asks for a square left of square 1, which does not exist.

Problem 8. Three pennies never reach the star two rows up (complete search over every placement). Argument: with three pennies at most two jumps are possible, since each jump removes one penny and one must remain. The last jump onto the star comes from row 0 over a penny in row 1 (a sideways jump within row 2 would need two pennies already in row 2, costing even more). So just before it, one penny is in row 1 and one in row 0 under it. The penny in row 1 arrived by jumping over the penny in row 0, which removed it; so row 0 must be refilled by another jump: three jumps at least, hence four pennies at least. With four pennies exactly one shape works, up to mirror image: the L, three pennies in row 0 starting from the star’s column and running to the right (or to the left), and one penny in row –1 under the first. Jumps: (1) the penny in row –1 jumps up over the penny in row 0 into row 1; (2) the penny at the far end of row 0 jumps sideways over its neighbour into the star’s column; (3) that penny jumps up over the penny in row 1 onto the star. Any other four-penny placement fails (search); children may test whatever they propose.

Problem 9. The army: row 0 at columns –2, –1, 0, 1, 2 and row –1 at columns 0, 1, 2 (the circles on board Q). The seven jumps: (1) row –1, column 0 jumps up over row 0 into row 1; (2) row 0, column –2 jumps right over column –1 into column 0; (3) row 0, column 0 jumps up over row 1 into row 2; (4) row –1, column 1 jumps up over row 0 into row 1; (5) row –1, column 2 jumps up over row 0 into row 1; (6) row 1, column 2 jumps left over column 1 into column 0; (7) row 1, column 0 jumps up over row 2 onto the star. The structure: the first three jumps are the L of Problem 8 getting a penny to row 2; jumps 4–6 bring a second penny to row 1 under it using two stacked pairs; jump 7 finishes. Seven pennies can never reach row 3 (the grades 4–5 table proves it with prices; for this table say “the price argument shows it, ask the 4–5 table at the share”).



Left: the eight-penny army of Problem 9 (board Q). Right: the second eight-penny army of reserve R3, which needs three rows below the line.

5.2 Hints, in order

- Problems 1–2: “Which pennies can jump at all? Count them.” “Is there a choice anywhere?”
- Problem 3: “On strip G, try the other first jump and play to the end. Which first jump was better?” “On strip F, what happens on the first jump no matter what?”
- Problem 4: “Before adding, guess the third column. Now check it.” “What does one jump do to the number of pennies?”
- Problem 5: “From the start there are two jumps. Draw both results, then work on each one.” “Are the two endings the same or mirror images?”
- Problem 6: “Which penny is the only one that can move?”
- Problem 7: “Try a big army, a tiny army, an army with a gap. Which square did the best one reach?” Then lead the price argument (Section 5.1, Problem 7) at the table with paper.
- Problem 8: “How many jumps can three pennies make?” “What must be under the star just before the last jump, and how did it get there?”
- Problem 9: “First get a penny two rows up: that is Problem 8. Then you need a penny just above the line in the star’s column while the one two rows up is still there. Where can that penny come from?”

5.3 If the table stalls

The mechanics are quick for third graders; stalls come from “fewest” searches. Set a limit: three plays per strip, then record and move on; the map (Problem 5) is where certainty comes from. If Problem 7 becomes random poking after four armies, stop and do the price argument at the table. If Problem 9 stalls after six minutes, give jumps 1–3 (the L) and let them find 4–7.

5.4 If the table finishes early (reserves, no printing needed)

- R1. Strip J again: find plays that end with 1, with 2 and with 4 pennies, then try to end with exactly 3. Three is impossible (the four stuck positions of the game have 1, 1, 2, 4 pennies).
- R2. Last jump wins on strip G (XX.XXX): two players alternate single jumps; whoever makes the last possible jump wins. The starter wins with 1 over 2 into 3: the rest is forced (4 over 3 into 2, then 6 over 5 into 4), three jumps, the starter makes the last. The other first jump, 5 over 4 into 3, loses: the reply 2 over 3 into 4 ends the game after two jumps.
- R3. A second army of eight for row 3 that is not a mirror image of Problem 9's (drawn above, at the right): row 0 at columns -2 to 2 , and column 0 at rows -1 , -2 , -3 . Jumps: (1) row -1 , column 0 jumps up over row 0 into row 1; (2) row 0, column -2 jumps right over column -1 into column 0; (3) row -3 , column 0 jumps up over row -2 into row -1 ; (4) row 0, column 0 jumps up over row 1 into row 2; (5) row 0, column 2 jumps left over column 1 into column 0; (6) row -1 , column 0 jumps up over row 0 into row 1; (7) row 1, column 0 jumps up over row 2 onto the star. These two are the only eight-penny armies for row 3 (up to mirror image; complete search). Board Q has only two rows below the line, so play this one on a spare grid.
- R4. Four holes, four different games. Outline a 3-wide, 4-tall board on a spare grid, 11 pennies. Hole in a corner: one penny is possible, and the lone survivor can only end on that corner or the other corner of the same short side (121 positions). Hole in the middle of a short side: only one jump exists in the whole game, then stuck with 10. Hole in the second square up a long side: one penny is possible and it can only end on the starting hole (154 positions). Hole on an inside square: two jumps, stuck with 9. A solution for the corner hole, naming squares (column, row) from the bottom-left corner (1,1): (1,3) over (1,2) into (1,1); (3,2) over (2,2) into (1,2); (1,1) over (1,2) into (1,3); (1,4) over (1,3) into (1,2); (2,4) over (2,3) into (2,2); (2,1) over (2,2) into (2,3); (3,4) over (3,3) into (3,2); (3,1) over (3,2) into (3,3); (3,3) over (2,3) into (1,3); (1,3) over (1,2) into (1,1).
- R5. Row 3 with seven pennies: let them try; it is impossible (the 4–5 table's price map: the seven most expensive squares below the line are worth $3 + 2 + 2 + 2 + 1 + 1 + 1 = 12$, and the star costs 13).

6. Grades 4–5 table (organizer; three children)

Order and timing (table work runs from 0:07 to 0:50, 43 minutes; the pages hold 77 minutes): Problem 1 by 0:14, Problem 3 by 0:22, Problem 4 by 0:33, Problem 5 by 0:39, Problem 6 by 0:50 (the second army of Problem 6 can be left for the share); Problems 7–8 are for the fast pair and for the share (the organizer plays the 19 jumps at the share in any case). Problem 2 (the map) is done in its printed place only if the table is ahead of these times; otherwise the “how do you know” for strip B is asked orally (hint: list every play) and the map is done at the end. If the table is still slow, shorten Problem 1 to strips A and B and cut Problem 3 to strip G only, so that Problem 4 (the prices) and Problems 5–6 (rows 2 and 3) are reached. The three children can work as a trio on Problems 2, 6 and 8.

Boards L, M and Q on the student pages carry column letters A–I and row numbers from the bottom, and board P (the same grid as Q, without the letters) is named the same way here; on L and M the thick line lies between rows 4 and 5 and the star is at E7; on P and Q the line lies between rows 4 and 5 and the star is at E8. Board K and board N are small and unlettered; for them rows are named from the thick line (row 1 just above it, row 0 just below it, then -1) and columns from the star's column (0).

6.1 Answers and arguments

Problem 1. Strip A (X.XXXX): the first jump is forced (4 over 3 into 2); afterwards every play ends with exactly one penny, on square 2 or on square 5 (7 positions, endings .X... andX.). Example: 4 over 3 into 2, 1 over 2 into 3, 6 over 5 into 4, then 3 over 4 into 5 or 4 over 3 into 2. Strip B (XX.XXX): starting with 1 over 2 into 3 forces 4 over 3 into 2 and 6 over 5 into 4, ending .X.X. with 2; starting with 5 over 4 into 3 forces 2 over 3 into 4, ending X.X.X with 3. Fewest 2, decided by the first jump. Strip C (XXXXX.): forced, 4 over 5 into 6, 2 over 3 into 4, X.X.X, 3 left. Number boxes: 1, 2, 3.

Problem 2. The map of XX.XXX: 6 positions, 5 arrows. XX.XXX $\rightarrow$.XXXX and XXX.X; .XXXX $\rightarrow$.X.XX; .X.XX $\rightarrow$.X.X. (stuck, 2); XXX.X $\rightarrow$ X.X.X (stuck, 3). No position is reached twice. The map proves that 2 is the fewest on strip B and that only the first jump matters. (If a child asks about strip A: its map has 7 positions and 7 arrows, and two different routes meet at .XX.; see reserve R1.)

Problem 3. Strips D, E, F: always exactly one square past the line, forced play. D: 2 over 3 into 4. E: 3 over 4 into 5, 1 over 2 into 3. F: 4 over 5 into 6, 2 over 3 into 4. Number boxes 1, 1, 1. Strip G: no army on squares 1 to 6 ever reaches square 8; square 7 is the farthest (all 63 armies checked by computer; the children cannot check all 63 by hand in the time, which is the point of Problem 4).

Problem 4. Full army on squares 1 to 6: total 20 before any jump. A jump to the right lands on a square whose price is exactly the sum of the two prices it used ($1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$, $3 + 5 = 8$, $5 + 8 = 13$, $8 + 13 = 21$), so the total stays the same. A jump to the left lands on a square that costs less than the two it used together (for 3 over 2 into 1 the change is $1 - 2 - 1 = -2$; for a jump from square k over $k - 1$ into $k - 2$ it is always negative), so the total goes down. Hence the total never goes up, and the largest total in any table is 20 (every play from the full army has at most 5 jumps; the tables have room for that). The star, square 8, costs 21, and a penny sitting on it would alone be worth 21: so no penny ever lands on square 8, whatever the play, and this covers every army at once, since any army on squares 1 to 6 is worth at most 20. Strip J (squares 1, 2, 4, 5, 6): total 18, same conclusion; possible jumps from the start are 1 over 2 into 3 (total stays 18), 5 over 4 into 3 (total drops to $1 + 1 + 2 + 8 = 12$) and 5 over 6 into 7 (stays 18). Small table: 1 to 3: 4 and 5; 1 to 4: 7 and 8; 1 to 5: 12 and 13; 1 to 6: 20 and 21; 1 to 7: 33 and 34; difference always 1. Why the next row will also give 1 without adding: adding one more penny adds the next price to the army; the old army was worth one less than its old target; old target plus the new penny's price is the new target, because each price is the sum of the two before it; so the new army is again worth one less than its new target. (Fibonacci identity: $1 + 1 + 2 + \cdots + F(n) = F(n + 2) - 1$.)

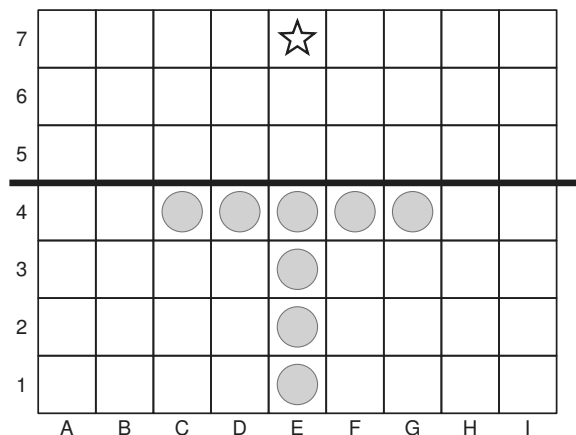
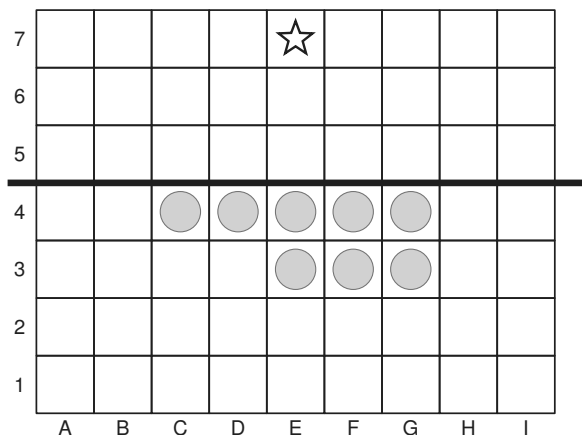
Problem 5. Three pennies never reach row 2 (complete search over every placement). Argument: at most two jumps with three pennies; the last jump onto the star comes from row 0 over row 1; the penny in row 1 got there by a jump that removed the penny in row 0 beneath it; refilling row 0 is a third jump; so four pennies at least. With four, the only shape up to mirror image is the L: row 0 at columns 0, 1, 2 (or 0, -1 , -2) and row -1 at column 0. Jumps: (0, -1) up over (0, 0) into (0, 1); (2, 0) left over (1, 0) into (0, 0); (0, 0) up over (0, 1) onto (0, 2). The row-2 price board of Problem 7 gives the clean proof: the three most expensive squares below the line are worth $3 + 2 + 2 = 7$ and the star costs 8.

Problem 6. Exactly two armies of eight work, up to mirror image (complete search). In board L coordinates (letters A–I, rows 1–7, line between rows 4 and 5, star E7):

Army (a): C4, D4, E4, F4, G4, E3, F3, G3. Jumps: 1. E3 over E4 to E5; 2. C4 over D4 to E4; 3. E4 over E5 to E6; 4. F3 over F4 to F5; 5. G3 over G4 to G5; 6. G5 over F5 to E5; 7. E5 over E6 to E7.

Army (b): C4, D4, E4, F4, G4, E3, E2, E1. Jumps: 1. E3 over E4 to E5; 2. C4 over D4 to E4; 3. E1 over E2 to E3; 4. E4 over E5 to E6; 5. G4 over F4 to E4; 6. E3 over E4 to E5; 7. E5 over E6 to E7.

Structure of (a): jumps 1–3 are the L reaching row 2; jumps 4–6 use two stacked pairs to bring a penny to E5 while E6 is still occupied; jump 7 finishes. Structure of (b): the L again, then a column refills E4 and E5 from below. Mirror images (columns C and D swapped with G and F) count as the same army.

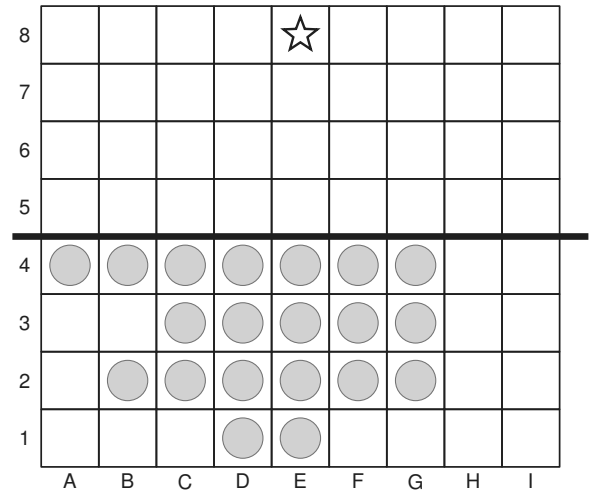
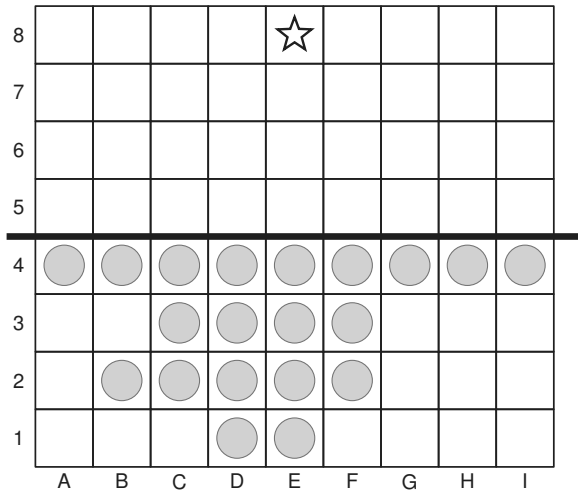


Army (a), left, and army (b), right, on board L. Army (a) is also the army printed on board M.

Problem 7. Board M prices: the star E7 costs 13; distance 1 costs 8, distance 2 costs 5, distance 3 costs 3, distance 4 costs 2, distance 5 or more costs 1. Army (a) is worth $1 + 2 + 3 + 2 + 1 + 2 + 1 + 1 = 13$ (C4 1, D4 2, E4 3, F4 2, G4 1, E3 2, F3 1, G3 1). After each of the seven jumps the total is still 13: every jump in this solution lands on a square worth exactly the two it used (jump 1: $2 + 3 = 5$ at E5; jump 2: $1 + 2 = 3$ at E4; jump 3: $3 + 5 = 8$ at E6; jump 4: $1 + 2 = 3$ at F5; jump 5: $1 + 1 = 2$ at G5; jump 6: $2 + 3 = 5$ at E5; jump 7: $5 + 8 = 13$ at E7). A jump away from the star lowers the total: for example E4 over E3 down to E2 removes $3 + 2$ and adds 1, total 9; F4 over G4 right to H4 removes $2 + 1$ and adds 1, total 11. General fact: no jump ever raises the total (a jump straight towards the star onto a square worth 2 or more is neutral; every other jump loses), checked by computer for every jump on a large board. The seven most expensive squares below the line: E4 (3), D4, F4, E3 (2 each), and three squares worth 1 (for instance C4, G4, F3): $3 + 2 + 2 + 2 + 1 + 1 + 1 = 12$, less than 13. So no army of seven pennies, placed anywhere, can ever put a penny on the star: its total starts at 12 or less, never rises, and a penny on the star is worth 13 by itself. The best eight are worth exactly 13, so an eight-penny army must be worth exactly 13, must never make a wasteful jump, and must end with nothing else on the board, which is what armies (a) and (b) do. Board N: the star costs 8; below the line, the square just under the line in the star's column costs 3, its three neighbours below the line (left, right and below it) cost 2 each, and every other square costs 1: the best three total $3 + 2 + 2 = 7$, less than 8, so three pennies can never reach row 2; the L is worth $3 + 2 + 1 + 2 = 8$ exactly.

Problem 8. Table: 1 row: 2; 2 rows: 4; 3 rows: 8; 4 rows: 20; 5 rows: never, for any number of pennies. Most children guess 16 for row 4; the pattern breaks. With 16 pennies, or any number up to 18, it is impossible: with Conway's exact weights (Section 8) the 18 heaviest squares below the line weigh 0.9787 in total, less than the star's weight 1. Nineteen is also impossible: the only 19-penny armies heavy enough weigh exactly 1, there are 84 of them, and a computer search shows none reaches row 4. Twenty works, and only four different 20-penny armies work up to mirror image (search). Board Q's army: row 4 at A–I, row 3 at C, D, E, F, row 2 at B, C, D, E, F, row 1 at D, E. The 19 jumps: 1. C3 over C4 to C5; 2. D3 over D4 to D5; 3. D1 over D2 to D3; 4. B2 over C2 to D2; 5. D2 over D3 to D4; 6. E3 over E4 to E5; 7. E1 over E2 to E3; 8. A4 over B4 to C4; 9. C4 over D4 to E4; 10. E4 over E5 to E6; 11. G4 over F4 to E4; 12. F2 over F3 to F4; 13. E3 over E4 to E5; 14. I4 over H4 to G4; 15. G4 over F4 to E4; 16. E5 over E6 to E7; 17. C5 over D5 to E5; 18. E4 over E5 to E6; 19. E6 over E7 to E8. Structure: jumps 1–2 park two helpers at C5 and D5 (row 1); jumps 3–5 rebuild D4; jumps 6–10 push a penny to E6 (row 2) using the column

E1–E4 and the left half of row 4; jumps 11–15 refill E4 and push again to E5; jump 16 reaches E7 (row 3); jump 17 brings the helpers to E5; jumps 18–19 climb to the star. This army weighs exactly 1, so every one of its 19 jumps is straight towards the star and nothing is left over.



Left: the 20-penny army of board Q. Right: the second 20-penny army of reserve R5.

6.2 Hints, in order

- Problem 1: “On strip A, what is the only first jump? After it, where can the last penny possibly end up? Try to make it end somewhere else.” “On strip B, play both first jumps to the end.”
- Problem 2: “Two arrows leave the start. Follow each branch to the end. Did any position appear twice?”
- Problem 3: “Which penny is the only one that can move?” On strip G: “Try the full army, then leave square 3 empty, then use only squares 4, 5, 6.”
- Problem 4: “Before adding, guess whether the total went up, down, or stayed.” “Which square’s price equals the two before it? All of them?” “If the total can never go up, and it starts at 20, what is the most a single penny on the strip could ever be worth?”
- Problem 5: “How many jumps can three pennies make? How many jumps did the L need?”
- Problem 6: “You know how to reach row 2 with four. What must sit at E5 when a penny is at E6, and where can it come from?” Then “two stacked pairs make a sideways jump into column E possible”. For the second army: “Instead of two pairs, could a column under E4 do the refilling?”
- Problem 7: “Which jumps kept the total? Which lowered it? Is there any jump that raised it?” “Add up the seven best squares below the line before you compare.”
- Problem 8: after five minutes with 16, say that 16 cannot work, and neither can 17, 18 or 19 (reasons in Section 6.1), then hand them board Q. Hint for the 19 jumps: “First park two pennies at C5 and D5; they are needed at the very end.” Second hint: “Get a penny to E6 the way the L does, using the column E1 to E4 and the left half of row 4.”

6.3 If the table stalls

Fourth and fifth graders stall on “find the army” problems (6 and 8), not on strips. Give Problem 6 eight minutes; then show army (a)’s first three jumps and let them finish; then give army (b) as the second. Problem 8’s 16-penny attempt is meant to fail: stop it after five minutes and go to board Q with the first hint. If the whole table is behind at 0:34, skip Problem 4’s second table and the small table and go straight to Problems 5 and 6, so that the row-2 and row-3 armies exist for the share.

6.4 If the table finishes early (reserves, no printing needed)

- R1. The map of strip A (X .XXXX) on scrap paper: 7 positions, 7 arrows: X .XXXX $\rightarrow$ XX . .XX; XX . .XX $\rightarrow$. .X .XX and XX .X . .; both of these $\rightarrow$. .XX . .; . .XX . . $\rightarrow$ X . and .X (stuck, 1 penny each). Two routes meet, so the map is not a tree; the endings are exactly squares 2 and 5.
- R2. Finish on the hole. Outline a 3-wide, 4-tall board on a spare grid, 11 pennies, hole in the bottom-left corner; finish with one penny on that corner. Solution, squares named (column, row) from (1,1) at the bottom-left: (1,3) over (1,2) into (1,1); (3,2) over (2,2) into (1,2); (1,1) over (1,2) into (1,3); (1,4) over (1,3) into (1,2); (2,4) over (2,3) into (2,2); (2,1) over (2,2) into (2,3); (3,4) over (3,3) into (3,2); (3,1) over (3,2) into (3,3); (3,3) over (2,3) into (1,3); (1,3) over (1,2) into (1,1). Replacing the last jump by (1,2) over (1,3) into (1,4) finishes on the other corner; these two corners are the only possible final squares (121 positions).
- R3. Row 5 on paper. Prices by distance from a star five rows up: 89, 55, 34, 21, 13, 8, 5, 3, 2, 1, 1, The 9-wide, 4-deep block of 36 squares under the star is worth 72, less than 89. The 25 squares at distance 5 to 9 are worth 61 and every further square is worth 1, so at least 53 pennies are needed before their prices could even add up to 89: no army of 52 or fewer can reach row 5. (The exact weights in Section 8 show that no finite army can.)
- R4. Last jump wins (two players alternate single jumps; the last jump wins). On XX .XX every play has exactly 2 jumps, so the second player always wins; on X .XXXX every play has exactly 4 jumps, second player again; on XXX .XXX every play has exactly 3 jumps, so the starter always wins whatever anyone does: the count “jumps = pennies removed” decides these games. On XX .XXX the starter wins only with 1 over 2 into 3.
- R5. A second 20-penny army for row 4 (drawn above, at the right), on a spare grid marked like board Q (9 columns A–I, 8 rows, line between rows 4 and 5, star at E8): row 4 at A, B, C, D, E, F, G; row 3 at C, D, E, F, G; row 2 at B, C, D, E, F, G; row 1 at D, E. Its 19 jumps (verified by replay): 1. C3 over C4 to C5; 2. D3 over D4 to D5; 3. D1 over D2 to D3; 4. B2 over C2 to D2; 5. D2 over D3 to D4; 6. E3 over E4 to E5; 7. E1 over E2 to E3; 8. G2 over F2 to E2; 9. E2 over E3 to E4; 10. F3 over F4 to F5; 11. G3 over G4 to G5; 12. A4 over B4 to C4; 13. C4 over C5 to C6; 14. D4 over D5 to D6; 15. E4 over E5 to E6; 16. G5 over F5 to E5; 17. E5 over E6 to E7; 18. C6 over D6 to E6; 19. E6 over E7 to E8. Structure: helpers parked at C5 and D5, the centre column rebuilt, a penny pushed to E7, then the helpers climb to C6 and D6 and jump across to E6 for the final climb. The search finds exactly four 20-penny armies up to mirror image (this one, board Q’s, and two others not printed); all four weigh exactly 1, so none of their solutions ever makes a jump that is not straight towards the star.

7. The share (8 minutes, at the launch board)

1. K–1 (2 min). One child builds the L on the launch board with the star two rows up and makes the three jumps. Another child answers “how many checkers for one star, two stars, three stars?” (2, 4, 6) and says why each star needs two: the one that lands and the one thrown away for it.
2. Grades 2–3 (2 min). One child places the eight-penny army (star three rows up) and makes the seven jumps. Another states the strip result: on a strip, an army never gets more than one square past the line, however big it is, and says the price reason in one sentence (“the total price never goes up, and the square two past the line costs more than the whole army”).
3. Grades 4–5 (3 min). One child shows the small table of Problem 4 (4 and 5, 7 and 8, $\dots$, 33 and 34: the army is always worth one less than its target). Another gives the table of fewest: 2, 4, 8 and then 20, not 16. The organizer states Conway’s theorem: no army of any size ever reaches row 5, and says that the proof gives each square a weight so that the star alone weighs as much as the whole infinite army; then, time permitting, plays the 19 jumps of board Q on the launch board (star four rows up).
4. Close (1 min). “One rule, three questions all hour: can it be done, how few, and how do you know for sure. The strongest answers today were the map, which lists everything, and the prices, which never go up.”

8. Mathematics behind the session (for the two mathematicians)

The object. A position is a finite set of occupied cells of a strip or of the square grid; a move replaces cells a, b, c in a line (a, b occupied, c empty) by c alone. Every problem on the pages is one of five questions about this dynamical system: existence (can a target position be reached), impossibility, optimality (fewest pegs left; fewest pegs used), completeness (which end positions or end squares occur), and counting (number of positions, number of first moves). Impossibility is where the mathematics is: the children meet three kinds of certificate.

1. Exhaustion: the map of positions (grades 2–3 Problem 5, grades 4–5 Problem 2). Complete for strips of 5 or 6 (5 or 6 positions) and for the 3×3 board (16). Beyond about 30 positions it is a computer’s job: the 4×4 board with a corner hole has 1687 positions, the 5×5 board with a centre hole 1,183,924.

2. Invariants and monovariants (potential functions). A function on positions that never increases under a move bounds what can be reached. The session uses four:

- Jumps + pegs left = starting pegs (grades 2–3 Problem 4): an invariant, because every jump removes exactly one peg. It settles two-player “last jump wins” games whenever every play has the same length (reserve R4 for grades 4–5).
- Fibonacci prices on the strip (grades 4–5 Problem 4): price $F(i)$ on square i with $F = 1, 1, 2, 3, 5, 8, \dots$. Since $F(i+2) = F(i+1) + F(i)$, a rightward jump is neutral and a leftward one loses. An army on squares 1 to k is worth $F(k+2) - 1$, one less than the target two past the line: the identity $F(1) + \dots + F(k) = F(k+2) - 1$. This is the one-dimensional shadow of Conway’s argument, sharp in both directions: one past the line is always reached (forced play), two never.
- Integer price maps on the grid (grades 4–5 Problem 7): price $F(K-d)$ at Manhattan distance d from the star, with 1 beyond. A jump straight towards the star is neutral (three consecutive Fibonacci numbers), a jump across the star’s column or row loses the price of the middle cell, a jump away

loses. Because along any line of three cells the distance pattern is $d, d+1, d+2$ (or reversed) or $d+1, d, d+1$, no jump raises the total (checked by machine on a large board). Consequences: row 2 needs weight 8 and the best three cells are worth 7; row 3 needs 13 and the best seven are worth 12; row 4 needs 34 ($K=9$) and the best eighteen are worth 33, so the integer map proves only that 18 pegs are too few for row 4.

- Conway's weights (*Winning Ways*, chapter 23; Aigner's "Moving into the desert with Fibonacci"): weight σ^d at distance d from the target, with $\sigma = (\sqrt{5} - 1)/2$, so $\sigma^2 + \sigma = 1$ and a jump towards the target is exactly neutral ($\sigma^{d+2} + \sigma^{d+1} = \sigma^d$), every other jump strictly loses. For a target in row 5 the whole half-plane below the line weighs exactly 1: a row at vertical distance k weighs $\sigma^k + 2\sigma^{k+1}/(1 - \sigma) = \sigma^k + 2\sigma^{k-1}$, and summing over $k \geq 5$ gives $\sigma^3 + 2\sigma^2 = (2\sigma - 1) + (2 - 2\sigma) = 1$. A finite army therefore weighs strictly less than 1, the weight never increases, and a peg on the target alone weighs 1: row 5 is unreachable by any finite army. For a target in row 4 the half-plane weighs $1/\sigma \approx 1.618$, so the argument gives only a lower bound: the 18 heaviest cells weigh (in exact form $-12 + 21\sigma) \approx 0.979 < 1$, so 19 pegs are necessary; the 19 heaviest weigh exactly 1, which forces a 19-peg army to be one of the 84 maximal ones and to waste nothing; our search shows all 84 fail. Twenty suffice: four armies up to mirror image, and the printed one weighs exactly 1, so its solution uses only jumps straight towards the star. Minimality of 20 was proved by Bell, Hirschberg and Guerrero-García ("The minimum size required of a solitaire army", *INTEGERS* 7, 2007), in agreement with our search. Rows 1, 2, 3 cost 2, 4, 8 (unique army for row 2, exactly two for row 3, by search); the doubling is a coincidence that the fourth case breaks.

Pagoda functions. Berlekamp, Conway and Guy call any function p on the cells with $p(a) + p(b) \geq p(c)$ for every move (a over b into c) a pagoda function; the sum of p over the position is then a monovariant, and a target position of larger pagoda value than the start is unreachable. Conway's weights, the Fibonacci prices and the integer price maps are pagoda functions; so is the counting invariant with $p \equiv 1$ turned into "pegs removed = jumps". The children's "prices never go up" is exactly this idea. Pagoda functions were introduced to prove impossibility results on the 33-hole English board (see also Beasley, *The Ins and Outs of Peg Solitaire*).

3. Colouring invariants (not on this week's pages; the natural next session). Colour square i of a strip by $i \bmod 3$, or cell (c, r) of the grid by $(c+r) \bmod 3$ or by $(c-r) \bmod 3$. Every jump uses three consecutive cells of three different colours, so all three colour counts change by 1 at once and their parities move together. For X. XXXXXX (8, hole at 2) this proves the last peg can only sit on squares 1, 4, 7 (the truth is 4 or 7: necessary, not sufficient); for the 3×4 board with a corner hole the two diagonal colourings together allow exactly the two squares on which the survivor can end; for the 5×5 board with a centre hole all counts are 8, 8, 8, and one peg is impossible: a one-minute proof for a game with over a million positions.

Where it leads.

- Undergraduate: monovariants and invariants are the first chapter of any olympiad combinatorics course (Engel, *Problem-Solving Strategies*, chapter on the invariance principle) and of the combinatorics units that teach Conway's Soldiers (Aigner's article; *Winning Ways*). The Fibonacci identity in Problem 4 and the exact arithmetic in $\mathbb{Z}[\sigma]$ (write every weight as $a + b\sigma$ with integers a, b) are good exercises.
- Peg solitaire on graphs (Beeler and Hoilman, "Peg solitaire on graphs", *Discrete Mathematics* 311, 2011): which graphs, from which starting hole, can be reduced to one peg. For paths our search up to 16 squares shows: $n=3$ yes (hole at an end); every even $n \geq 4$ yes, with the hole at the 2nd square from an end (and for $n \geq 6$ also the 5th); every odd $n \geq 5$ no, from every hole. Beeler and Hoilman settle paths in general (and other families such as complete and complete bipartite graphs), and the subject has a growing literature on which graphs are solvable.

- Computation: deciding whether a general peg-solitaire position can be reduced to one peg is NP-complete (Uehara and Iwata, “Generalized Hi-Q is NP-complete”, 1990), so certificates like pagoda functions, which are checked by one addition per move, are the tool that scales.
- Research directions the children’s questions touch: minimal armies for other targets and other boards (hexagonal and triangular grids, higher dimensions), and the fact that with infinitely many moves allowed, row 5 becomes reachable (Tatham and Taylor, “Reaching row five in solitaire army”), which shows exactly what finiteness does in Conway’s proof.

Verification note. Every position count, ending, minimal army and move sequence in this guide was checked by exhaustive search (`verify_curator.py` and the bank’s `verify_*.py`, kept in the working directory); the row-4 results (no 19-peg army works; four 20-peg armies work) come from a bitboard search of every army heavy enough to reach the target, and the weights were computed exactly in $\mathbb{Z}[\sigma]$.